Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

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	Budding yeast	Nematode	Fruit fly	Mouse
C008 **TOR** 4 genes / 4 species 6 direct pairs	**Y1 TOR1** GenAge 679 **Reviewed example** Deletion **Replicative life ↑** PMID 16293764	**W1 let-363** GenAge 32 **Reviewed example** RNAi **Adult life ↑** PMID 14668850	**F1 Tor** GenAge 678 **Reviewed example** Dominant-negative **Adult life ↑** PMID 15186745	**M1 Mtor** GenAge 1809 **Reviewed example** Hypomorph **Median survival ↑** PMID 23994476

Direct pairs: W1–F1 (1:1); W1–M1 (1:1); W1–Y1 (1:m); F1–M1 (1:1); F1–Y1 (1:m); M1–Y1 (1:m)

Yeast replicative lifespan differs from animal survival. Yeast TOR1 pairs are non-1:1; TOR and S6K share a pathway.

C073 **S6 kinase** 3 genes / 3 species 2 direct pairs	No gene in this candidate network	**W1 rsks-1** GenAge 578 **Reviewed example** RNAi **Mean life ↑** PMID 17266680	**F1 S6k** GenAge 583 **Reviewed example** Dominant-negative **Adult life ↑** PMID 15186745	**M1 Rps6kb1** GenAge 1737 **Reviewed example** Global knockout **Female life ↑*** PMID 19797661

Direct pairs: W1–F1 (1:1); F1–M1 (1:1)

* Mouse benefit is female-specific; male increase was not significant. The 2011 correction concerns a worm strain, not mouse survival. No worm–mouse pair is implied.

C019 **Insulin/IGF receptor** 4 genes / 3 species 3 direct pairs	No gene in this candidate network	**W1 daf-2** GenAge 163 **Reviewed example** Hypomorph **Adult life ↑** PMID 8247153	**F1 InR** GenAge 368 **Reviewed example** Hypomorph **Female life ↑** PMID 11292875	**M1 Igf1r** GenAge 365 **Reviewed example** Heterozygous KO **Female life ↑*†** PMID 12483226 **M2 Insr** GenAge 373 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12543978, 15713666

Direct pairs: W1–F1 (1:1); W1–M1 (1:n); W1–M2 (1:n)

* Original female Igf1r benefit; male increase not significant. † Not reproduced in a later cohort (PMID 22132081).

Insr is a separate paralog; no fly–mouse pair is implied.

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.

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	Budding yeast	Nematode	Fruit fly	Mouse
C001 **AMPK** 3 genes / 3 species 2 direct pairs	**Y1 SNF1** GenAge 619 **Reviewed example** Overexpression **Replicative life ↓** PMID 10921902	**W1 aak-2** GenAge 1 **Reviewed example** Extra gene copies **Mean life ↑** PMID 15574588	**F1 SNF1A** GenAge 1846 **Reviewed example** Adult tissue expression **Adult life ↑** PMID 23312286	No gene in this candidate network

Direct pairs: W1–F1 (1:1); F1–Y1 (1:1)

Yeast result is replicative lifespan; worm/fly interventions differ. No direct yeast–worm pair is implied.

C004 **ATG7** 3 genes / 3 species 3 direct pairs	**Y1 ATG7** GenAge 1218 **Reviewed example** Deletion **Chronological life ↓** PMID 19302372	**W1 atg-7** GenAge 23 **Reviewed example** RNAi in eat-2 **DR benefit reduced‡** PMID 17912023	**F1 Atg7** GenAge 21 **Reviewed example** Null mutation **Adult life ↓** PMID 18056421	No gene in this candidate network

Direct pairs: W1–F1 (1:1); W1–Y1 (1:1); F1–Y1 (1:1)

‡ Worm cell: dietary-restricted eat-2 mutants, not a general wild-type result. GenAge labels differ from this selected experiment; timing and background matter.

C080 **SOD2** 4 genes / 4 species 6 direct pairs	**Y1 SOD2** GenAge 623 **Reviewed example** Deletion in sch9Δ **CLS benefit lost§** PMID 12586694	**W1 sod-2** GenAge 985 **Reviewed example** Deletion **Adult life ↑** PMID 19197346	**F1 Sod2** GenAge 624 **Reviewed example** RNAi **Adult life ↓** PMID 12456885	**M1 Sod2** GenAge 827 **Reviewed example** Connective-tissue KO **Survival ↓** PMID 21108731

Direct pairs: W1–F1 (m:1); W1–M1 (m:1); W1–Y1 (n:1); F1–M1 (1:1); F1–Y1 (1:1); M1–Y1 (1:1)

§ Yeast: loss of sch9Δ chronological lifespan extension. Mouse: connective-tissue deficiency. Worm deletion effects depend on genetic background.

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.

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GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.

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	Budding yeast	Nematode	Fruit fly	Mouse
C007 **TRX1** 5 genes / 4 species 5 direct pairs	**Y1 TRX1** GenAge 1286 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943, 20883526	**W1 trx-1** GenAge 31 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16324156	**F1 Trx-2** GenAge 685 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17567437 **F2 TrxT** GenAge 686 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 17301052	**M1 Txn1** GenAge 691 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 12230882, 21873593
Direct pairs: W1–M1 (m:1); W1–Y1 (n:m); F2–M1 (1:1); F1–Y1 (n:m); M1–Y1 (1:m)				
C064 **TSA1** 4 genes / 4 species 3 direct pairs	**Y1 TSA1** GenAge 2109 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 27264606	**W1 prdx-2** GenAge 516 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16673436	**F1 Jafrac1** GenAge 856 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 19720829, 20976250	**M1 Prdx1** GenAge 518 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 12891360
Direct pairs: W1–Y1 (1:m); F1–Y1 (n:m); M1–Y1 (n:m)				
C077 **AAC3** 4 genes / 3 species 5 direct pairs	**Y1 AAC3** GenAge 1553 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043 **Y2 PET9** GenAge 1595 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18809618	**W1 ant-1.1** GenAge 666 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836	No gene in this candidate network	**M1 Slc25a4** GenAge 615 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 10220377
Direct pairs: W1–M1 (m:n); W1–Y1 (n:m); W1–Y2 (n:m); M1–Y1 (n:m); M1–Y2 (n:m)				

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	Budding yeast	Nematode	Fruit fly	Mouse
C002 **akt-1** 4 genes / 3 species 5 direct pairs	No gene in this candidate network	**W1 akt-1** GenAge 10 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808, 23935948	**F1 Akt1** GenAge 832 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 20976250	**M1 Akt1** GenAge 1810 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 23935948 **M2 Akt2** GenAge 2231 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 28681509
Direct pairs: W1–F1 (m:1); W1–M1 (m:n); W1–M2 (m:n); F1–M1 (1:n); F1–M2 (1:n)				
C104 **ATG5** 3 genes / 3 species 2 direct pairs	**Y1 ATG5** GenAge 1305 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 atg-5** GenAge 897 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19469880	No gene in this candidate network	**M1 Atg5** GenAge 1862 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 23939249
Direct pairs: W1–M1 (1:1); M1–Y1 (1:1)				
C005 **ATG8** 3 genes / 3 species 3 direct pairs	**Y1 ATG8** GenAge 1306 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 lgg-1** GenAge 955 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18219227, 19469880	**F1 Atg8a** GenAge 22 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 17617737, 18059160 19550147	No gene in this candidate network
Direct pairs: W1–F1 (1:1); W1–Y1 (1:1); F1–Y1 (n:1)				

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	Budding yeast	Nematode	Fruit fly	Mouse
C034 **BMH1** 5 genes / 3 species 6 direct pairs	**Y1 BMH1** GenAge 1033 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 19805817, 20657825	**W1 ftt-2** GenAge 288 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 16860373, 18423931	**F1 14-3-3epsilon** GenAge 831 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18665908	No gene in this candidate network
	Y2 BMH2 GenAge 1670 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 19805817	**W2 par-5** GenAge 493 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 16860373, 18423931		
Direct pairs: W1–Y1 (n:m); W1–Y2 (n:m); W2–Y1 (n:m); W2–Y2 (n:m); F1–Y1 (1:m); F1–Y2 (1:m)				
C017 **CAT5** 3 genes / 3 species 3 direct pairs	**Y1 CAT5** GenAge 1436 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 17492370, 18340043	**W1 clk-1** GenAge 130 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 8638122, 10202142 12709403, 15269213	No gene in this candidate network	**M1 Coq7** GenAge 136 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16195414, 25106098
Direct pairs: W1–M1 (1:1); W1–Y1 (1:1); M1–Y1 (1:1)				
C012 **CTA1** 3 genes / 3 species 3 direct pairs	**Y1 CTA1** GenAge 1694 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 20696905	No gene in this candidate network	**F1 Cat** GenAge 91 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 8108730	**M1 Cat** GenAge 92 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 15879174, 16144468
Direct pairs: F1–M1 (1:1); F1–Y1 (n:1); M1–Y1 (1:1)				

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	Budding yeast	Nematode	Fruit fly	Mouse
C046 **KSS1** 4 genes / 3 species 3 direct pairs	**Y1 KSS1** GenAge 397 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 17662940	**W1 mpk-1** GenAge 961 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18059442, 20624915	**F1 Mpk2** GenAge 1832 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 22014527 **F2 p38b** GenAge 1831 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 22014527	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); F1–Y1 (n:m); F2–Y1 (n:m)				
C099 **MNS1** 3 genes / 3 species 2 direct pairs	**Y1 MNS1** GenAge 1513 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20657825	**W1 D2030.1** GenAge 912 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19302370	**F1 alpha-Man-I** GenAge 833 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19302370	No gene in this candidate network
Direct pairs: W1–F1 (1:1); F1–Y1 (n:1)				
C044 **nac-2** 4 genes / 3 species 3 direct pairs	No gene in this candidate network	**W1 nac-2** GenAge 452 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 14678010 **W2 nac-3** GenAge 453 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12480943	**F1 Indy** GenAge 366 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11118146, 17571923 18460334	**M1 Slc13a1** GenAge 825 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 21651971
Direct pairs: W1–F1 (m:1); W2–F1 (m:1); F1–M1 (m:n)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
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	Budding yeast	Nematode	Fruit fly	Mouse
C084 **QCR7** 3 genes / 3 species 2 direct pairs	**Y1 QCR7** GenAge 1130 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043, 20657825	**W1 T02H6.11** GenAge 639 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12447374, 20688098	**F1 CG17856** GenAge 838 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19747824	No gene in this candidate network
Direct pairs: W1–F1 (1:n); F1–Y1 (n:1)				
C045 **RIM11** 3 genes / 3 species 2 direct pairs	**Y1 RIM11** GenAge 1320 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 jnk-1** GenAge 381 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 15767565, 18832074	**F1 bsk** GenAge 837 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 20976250	No gene in this candidate network
Direct pairs: W1–F1 (1:1); F1–Y1 (n:1)				
C079 **SOD1** 3 genes / 3 species 3 direct pairs	**Y1 SOD1** GenAge 621 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 12586694, 15659212 17460215	**W1 sod-1** GenAge 984 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18073424	**F1 Sod** GenAge 622 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 2539600, 8108730 9858546	No gene in this candidate network
Direct pairs: W1–F1 (1:1); W1–Y1 (n:1); F1–Y1 (1:1)				
C016 **YKU70** 3 genes / 3 species 2 direct pairs	**Y1 YKU70** GenAge 776 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 10521401, 20657825	**W1 cku-70** GenAge 129 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16099946	No gene in this candidate network	**M1 Xrcc6** GenAge 736 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17875923
Direct pairs: W1–M1 (1:1); M1–Y1 (1:1)				

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C023 **acdh-13** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 acdh-13** GenAge 300 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836	**F1 Egm** GenAge 215 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16434470	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C095 **AFG3** 2 genes / 2 species 1 direct pair	**Y1 AFG3** GenAge 1650 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043	**W1 spg-7** GenAge 753 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17411345	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C100 **atg-3** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 atg-3** GenAge 894 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19469880	**F1 Aut1** GenAge 835 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18219227	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C038 **ATP1** 2 genes / 2 species 1 direct pair	**Y1 ATP1** GenAge 1604 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 17492370, 18340043	**W1 H28O16.1** GenAge 331 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17411345	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

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	Budding yeast	Nematode	Fruit fly	Mouse
C006 **ATP2** 2 genes / 2 species 1 direct pair	**Y1 ATP2** GenAge 29 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 12242224, 18340043	**W1 atp-2** GenAge 25 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11410594, 12471266 17411345, 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C009 **BCY1** 2 genes / 2 species 1 direct pair	**Y1 BCY1** GenAge 42 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 8195187, 20421943	No gene in this candidate network	No gene in this candidate network	**M1 Prkar2b** GenAge 1738 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19536287
Direct pairs: M1–Y1 (n:1)				
C010 **bub-3** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 bub-3** GenAge 900 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18059442	No gene in this candidate network	**M1 Bub3** GenAge 51 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16476774
Direct pairs: W1–M1 (1:1)				
C105 **CDH1** 2 genes / 2 species 1 direct pair	**Y1 CDH1** GenAge 1629 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 fzy-1** GenAge 941 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18059442	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C014 **cdk-7** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 cdk-7** GenAge 107 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	**M1 Cdk7** GenAge 1813 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 22505032
Direct pairs: W1–M1 (1:1)				

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	Budding yeast	Nematode	Fruit fly	Mouse
C101 **CELE_C25H3.9** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 CELE_C25H3.9** GenAge 905 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20688098	**F1 l(3)neo18** GenAge 862 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19747824	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C082 **chn-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 chn-1** GenAge 1771 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 23144747	No gene in this candidate network	**M1 Stub1** GenAge 633 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18411298
Direct pairs: W1–M1 (1:1)				
C089 **COR1** 2 genes / 2 species 1 direct pair	**Y1 COR1** GenAge 1223 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 ucr-1** GenAge 697 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836, 20688098 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C021 **CYC1** 2 genes / 2 species 1 direct pair	**Y1 CYC1** GenAge 1045 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 19030232, 20657825 20883526	**W1 cyc-2.1** GenAge 200 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
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	Budding yeast	Nematode	Fruit fly	Mouse
C013 **CYC3** 3 genes / 2 species 2 direct pairs	**Y1 CYC3** GenAge 1414 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043 **Y2 CYT2** GenAge 1047 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043, 19030232 20657825	**W1 cchl-1** GenAge 94 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12447374, 16103914	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m); W1–Y2 (1:m)				
C123 **CYS4** 2 genes / 2 species 1 direct pair	**Y1 CYS4** GenAge 1569 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20883526	No gene in this candidate network	**F1 Cbs** GenAge 1791 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 21930912	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				
C018 **CYT1** 2 genes / 2 species 1 direct pair	**Y1 CYT1** GenAge 1701 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043	**W1 cyc-1** GenAge 145 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16103914, 20688098	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C132 **dgk-5** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 dgk-5** GenAge 1931 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24889782	**F1 rdgA** GenAge 1929 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24889782	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

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	Budding yeast	Nematode	Fruit fly	Mouse
C020 **DNM1** 2 genes / 2 species 1 direct pair	**Y1 DNM1** GenAge 185 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17173038	No gene in this candidate network	**F1 Drp1** GenAge 2215 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 28878259	No gene in this candidate network
Direct pairs: F1–Y1 (1:m)				
C098 **Dys** 2 genes / 2 species 1 direct pair	No gene in this candidate network	No gene in this candidate network	**F1 Dys** GenAge 843 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18221418	**M1 Dmd** GenAge 815 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17360850
Direct pairs: F1–M1 (1:n)				
C022 **eef-1A.1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 eef-1A.1** GenAge 2034 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	**F1 Ef1alpha48D** GenAge 208 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 2508089	No gene in this candidate network
Direct pairs: W1–F1 (m:n)				
C097 **EGD2** 2 genes / 2 species 1 direct pair	**Y1 EGD2** GenAge 1415 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 Y65B4BR.5** GenAge 765 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17411345	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C024 **ERO1** 2 genes / 2 species 1 direct pair	**Y1 ERO1** GenAge 1863 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24391512	**W1 ero-1** GenAge 225 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17411345	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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	Budding yeast	Nematode	Fruit fly	Mouse
C029 **F55B11.1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 F55B11.1** GenAge 265 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	**F1 ry** GenAge 582 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17435236	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C026 **FAT1** 2 genes / 2 species 1 direct pair	**Y1 FAT1** GenAge 1231 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 acs-20** GenAge 245 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:1)				
C032 **FBP1** 2 genes / 2 species 1 direct pair	**Y1 FBP1** GenAge 280 **GenAge annotation** **Anti-Longevity** Lifespan: Decrease Refs: 16199065	**W1 fbp-1** GenAge 279 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17673661	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C033 **frh-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 frh-1** GenAge 287 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 15771615, 16293572 23247094	**F1 fh** GenAge 848 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18258192	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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	Budding yeast	Nematode	Fruit fly	Mouse
C111 **GDH3** 2 genes / 2 species 1 direct pair	**Y1 GDH3** GenAge 1059 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20657825	**W1 gdh-1** GenAge 2043 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C042 **GLK1** 4 genes / 2 species 4 direct pairs	**Y1 GLK1** GenAge 1624 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 19030232, 24586198 **Y2 HXK2** GenAge 357 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11000115, 20421943	No gene in this candidate network	**F1 Hex-A** GenAge 2150 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 26378219 **F2 Hex-C** GenAge 2151 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 26378219	No gene in this candidate network
Direct pairs: F1–Y1 (n:m); F1–Y2 (n:m); F2–Y1 (n:m); F2–Y2 (n:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C035 **GPA2** 6 genes / 2 species 5 direct pairs	**Y1 GPA2** GenAge 319 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 11000115, 20421943	**W1 egl-30** GenAge 919 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19043554, 20647387 **W2 goa-1** GenAge 942 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19043554 **W3 gpa-3** GenAge 314 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17187771 **W4 gsa-1** GenAge 943 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18059442 **W5 odr-3** GenAge 481 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17187771, 20332814	No gene in this candidate network	No gene in this candidate network

Direct pairs: W1–Y1 (n:m); W2–Y1 (n:m); W3–Y1 (n:m); W4–Y1 (n:m); W5–Y1 (n:m)

C037 **GPX2** 2 genes / 2 species 1 direct pair	**Y1 GPX2** GenAge 1578 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 19030232	No gene in this candidate network	No gene in this candidate network	**M1 Gpx4** GenAge 324 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17895430

Direct pairs: M1–Y1 (n:m)

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C114 **GSY2** 2 genes / 2 species 1 direct pair	**Y1 GSY2** GenAge 1364 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526	No gene in this candidate network	**F1 GlyS** GenAge 1919 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 25059425	No gene in this candidate network
Direct pairs: F1–Y1 (1:m)				
C128 **GTR1** 2 genes / 2 species 1 direct pair	**Y1 GTR1** GenAge 1746 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 22912585	**W1 raga-1** GenAge 2062 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22560223	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C039 **hsp-4** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 hsp-4** GenAge 345 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16741121	**F1 Hsc70-3** GenAge 853 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18059160	No gene in this candidate network
Direct pairs: W1–F1 (m:1)				
C112 **HXT5** 2 genes / 2 species 1 direct pair	**Y1 HXT5** GenAge 1239 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 fgt-1** GenAge 1912 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24024580	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m)				
C028 **IDH2** 2 genes / 2 species 1 direct pair	**Y1 IDH2** GenAge 358 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16293764, 18340043 19030232	**W1 idha-1** GenAge 260 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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Complete visual catalogue • 320 genes • 206 direct pairs

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C031 **IDP1** 4 genes / 2 species 3 direct pairs	**Y1 IDP1** GenAge 1346 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043, 20421943 **Y2 IDP2** GenAge 1366 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043, 20883526 **Y3 IDP3** GenAge 1642 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043, 19030232	**W1 idh-1** GenAge 275 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W1–Y2 (n:m); W1–Y3 (n:m)				
C090 **INP53** 2 genes / 2 species 1 direct pair	**Y1 INP53** GenAge 1367 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 16293764, 18340043 20883526	**W1 unc-26** GenAge 703 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 9789046	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C106 **ire-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 ire-1** GenAge 948 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 20460307	**F1 Ire1** GenAge 2148 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 27783936	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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	Budding yeast	Nematode	Fruit fly	Mouse
C085 **KGD1** 2 genes / 2 species 1 direct pair	**Y1 KGD1** GenAge 1078 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20657825, 20883526	**W1 ogdh-1** GenAge 650 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C047 **LAT1** 2 genes / 2 species 1 direct pair	**Y1 LAT1** GenAge 400 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 17200108	**W1 dlat-1** GenAge 2030 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C048 **lbp-6** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 lbp-6** GenAge 402 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 16673436, 19764929	**F1 fabp** GenAge 1962 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 22997544	No gene in this candidate network
Direct pairs: W1–F1 (m:1)				
C049 **LPD1** 2 genes / 2 species 1 direct pair	**Y1 LPD1** GenAge 1245 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 dld-1** GenAge 413 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836, 23173729	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

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	Budding yeast	Nematode	Fruit fly	Mouse
C052 **mcm-2** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 mcm-2** GenAge 426 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 17411345, 18006689	No gene in this candidate network	**M1 Mcm2** GenAge 427 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17717065
Direct pairs: W1–M1 (1:1)				
C053 **MDH1** 3 genes / 2 species 2 direct pairs	**Y1 MDH1** GenAge 1615 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18381895 **Y2 MDH2** GenAge 1616 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18381895	**W1 mdh-2** GenAge 428 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m); W1–Y2 (1:m)				
C003 **MEP2** 3 genes / 2 species 2 direct pairs	**Y1 MEP2** GenAge 432 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16418483 **Y2 MEP3** GenAge 433 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16418483	**W1 amt-2** GenAge 12 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17608836	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W1–Y2 (n:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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	Budding yeast	Nematode	Fruit fly	Mouse
C054 **mev-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 mev-1** GenAge 435 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 9716135, 15269213	**F1 SdhC** GenAge 594 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17854771	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C117 **MRS4** 2 genes / 2 species 1 direct pair	**Y1 MRS4** GenAge 1439 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043, 19030232	**W1 mfn-1** GenAge 1750 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22253756	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C055 **MSH2** 2 genes / 2 species 1 direct pair	**Y1 MSH2** GenAge 1371 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526	No gene in this candidate network	No gene in this candidate network	**M1 Msh2** GenAge 444 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 12629043
Direct pairs: M1–Y1 (1:1)				
C056 **MXR1** 2 genes / 2 species 1 direct pair	**Y1 MXR1** GenAge 450 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 15141092, 19049972	No gene in this candidate network	No gene in this candidate network	**M1 Msra** GenAge 446 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 11606777, 19487311
Direct pairs: M1–Y1 (1:1)				
C050 **NAM2** 2 genes / 2 species 1 direct pair	**Y1 NAM2** GenAge 1259 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 lars-2** GenAge 417 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12447374	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
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	Budding yeast	Nematode	Fruit fly	Mouse
C131 **nsun-5** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 nsun-5** GenAge 2079 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 25635753	**F1 CG42358** GenAge 1854 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 25635753	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C058 **nuo-5** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 nuo-5** GenAge 480 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	**F1 ND75** GenAge 1898 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24243023	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C057 **pat-3** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 pat-3** GenAge 1786 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 23144747	**F1 mys** GenAge 451 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 14570233, 24400780	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C059 **pat-4** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 pat-4** GenAge 494 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16103914, 17411345 17608836	**F1 Ilk** GenAge 1910 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24400780	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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	Budding yeast	Nematode	Fruit fly	Mouse
C060 **pcm-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 pcm-1** GenAge 969 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18772467	**F1 Pcmt** GenAge 498 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 11742076, 18772467	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C061 **PHB1** 2 genes / 2 species 1 direct pair	**Y1 PHB1** GenAge 504 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 9259555, 12882345	**W1 phb-1** GenAge 970 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19812672	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C062 **PHB2** 2 genes / 2 species 1 direct pair	**Y1 PHB2** GenAge 505 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 9259555, 12882345 16710639	**W1 phb-2** GenAge 971 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19812672	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C015 **PHO85** 2 genes / 2 species 1 direct pair	**Y1 PHO85** GenAge 1317 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043, 20421943	No gene in this candidate network	**F1 Cdk5** GenAge 108 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17368005	No gene in this candidate network
Direct pairs: F1–Y1 (n:m)				
C120 **PIM1** 2 genes / 2 species 1 direct pair	**Y1 PIM1** GenAge 1467 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526	**W1 C34B2.6** GenAge 2028 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C110 **pme-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 pme-1** GenAge 1757 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 23870130	No gene in this candidate network	**M1 Parp1** GenAge 1010 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 20561897
Direct pairs: W1–M1 (1:1)				
C093 **pnc-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 pnc-1** GenAge 741 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17335870	**F1 Naam** GenAge 871 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18678867	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C115 **PTC5** 2 genes / 2 species 1 direct pair	**Y1 PTC5** GenAge 1373 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526	**W1 pdp-1** GenAge 2065 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 21533078	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C102 **Pten** 2 genes / 2 species 1 direct pair	No gene in this candidate network	No gene in this candidate network	**F1 Pten** GenAge 876 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 21111239	**M1 Pten** GenAge 1012 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 17400585, 22405073
Direct pairs: F1–M1 (1:1)				
C066 **RPD3** 2 genes / 2 species 1 direct pair	**Y1 RPD3** GenAge 555 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 10512855	No gene in this candidate network	**F1 Rpd3** GenAge 556 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12459580	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C069 **RPL10** 2 genes / 2 species 1 direct pair	**Y1 RPL10** GenAge 561 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17174052	**W1 rpl-10** GenAge 2047 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C121 **RPL17B** 2 genes / 2 species 1 direct pair	**Y1 RPL17B** GenAge 1503 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18423200	**W1 rpl-17** GenAge 2050 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C119 **RPL1B** 2 genes / 2 species 1 direct pair	**Y1 RPL1B** GenAge 1460 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18423200	**W1 rpl-1** GenAge 2042 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C126 **RPL23A** 2 genes / 2 species 1 direct pair	**Y1 RPL23A** GenAge 1725 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18423200	**W1 rpl-23** GenAge 2038 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
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	Budding yeast	Nematode	Fruit fly	Mouse
C070 **RPL31A** 2 genes / 2 species 1 direct pair	**Y1 RPL31A** GenAge 562 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 16293764, 17174052 18423200, 19030232 20883526	**W1 rpl-31** GenAge 2040 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C067 **RPL4A** 2 genes / 2 species 1 direct pair	**Y1 RPL4A** GenAge 1461 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043, 18423200	**W1 rpl-4** GenAge 559 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17266679	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C127 **RPL7A** 2 genes / 2 species 1 direct pair	**Y1 RPL7A** GenAge 1731 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18423200	**W1 rpl-7** GenAge 2048 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C122 **RPL8B** 2 genes / 2 species 1 direct pair	**Y1 RPL8B** GenAge 1563 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18423200	**W1 rpl-7A** GenAge 2055 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C068 **RPL9A** 3 genes / 2 species 2 direct pairs	**Y1 RPL9A** GenAge 1657 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043, 18423200 **Y2 RPL9B** GenAge 1658 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043, 18423200	**W1 rpl-9** GenAge 560 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17266679	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m); W1–Y2 (1:m)				
C133 **RPO31** 2 genes / 2 species 1 direct pair	**Y1 RPO31** GenAge 2240 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 29186112	**W1 rpc-1** GenAge 2239 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 29186112	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C116 **RPS0B** 2 genes / 2 species 1 direct pair	**Y1 RPS0B** GenAge 1422 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18423200	**W1 rps-0** GenAge 2035 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C071 **RPS11A** 2 genes / 2 species 1 direct pair	**Y1 RPS11A** GenAge 1606 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043, 18423200	**W1 rps-11** GenAge 566 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17266679, 17411345 22103665, 23935948	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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Complete visual catalogue • 320 genes • 206 direct pairs

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C129 **RPS12** 2 genes / 2 species 1 direct pair	**Y1 RPS12** GenAge 2126 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 26456335	**W1 rps-12** GenAge 1767 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 23144747	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C072 **RPS26B** 2 genes / 2 species 1 direct pair	**Y1 RPS26B** GenAge 1712 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043	**W1 rps-26** GenAge 570 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17266679	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C130 **RPT1** 2 genes / 2 species 1 direct pair	**Y1 RPT1** GenAge 2014 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 21931558	**W1 rpt-1** GenAge 1787 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 23144747	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C094 **RTC2** 3 genes / 2 species 2 direct pairs	**Y1 RTC2** GenAge 1274 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943 **Y2 YPQ1** GenAge 1748 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 22912585	**W1 Y43H11AL.2** GenAge 749 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m); W1–Y2 (1:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C027 **SAC1** 2 genes / 2 species 1 direct pair	**Y1 SAC1** GenAge 1501 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 sac-1** GenAge 248 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C074 **SAM1** 3 genes / 2 species 2 direct pairs	**Y1 SAM1** GenAge 1660 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043	**W1 sams-1** GenAge 584 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16103914 **W2 sams-3** GenAge 903 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W2–Y1 (n:m)				
C075 **SDH2** 2 genes / 2 species 1 direct pair	**Y1 SDH2** GenAge 2136 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 26456335	No gene in this candidate network	**F1 SdhB** GenAge 593 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17056719	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				
C030 **SIP2** 2 genes / 2 species 1 direct pair	**Y1 SIP2** GenAge 603 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 10921902, 19030232 20421943, 21906795	**W1 aakb-1** GenAge 269 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16673436	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C076 **Sirt6** 2 genes / 2 species 1 direct pair	No gene in this candidate network	No gene in this candidate network	**F1 Sirt6** GenAge 611 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17159295	**M1 Sirt6** GenAge 610 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 16439206, 22367546
Direct pairs: F1–M1 (1:1)				
C124 **SKM1** 2 genes / 2 species 1 direct pair	**Y1 SKM1** GenAge 1608 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 pak-1** GenAge 1897 **GenAge annotation** **Unannotated** Lifespan: Increase Refs: 23524941	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C078 **SNF4** 2 genes / 2 species 1 direct pair	**Y1 SNF4** GenAge 620 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 10921902	**W1 aakg-2** GenAge 1849 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 22737090	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:1)				
C081 **SPT4** 2 genes / 2 species 1 direct pair	**Y1 SPT4** GenAge 1716 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043	**W1 spt-4** GenAge 627 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C041 **SSA3** 3 genes / 2 species 2 direct pairs	**Y1 SSA3** GenAge 1538 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20883526 **Y2 SSA4** GenAge 1159 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943, 20657825	No gene in this candidate network	**F1 Hsp68** GenAge 351 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 14602080, 20976250	No gene in this candidate network
Direct pairs: F1–Y1 (n:m); F1–Y2 (n:m)				
C040 **SSC1** 2 genes / 2 species 1 direct pair	**Y1 SSC1** GenAge 2110 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 27346163	**W1 hsp-6** GenAge 346 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 11959102, 17189267	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C092 **SSO1** 3 genes / 2 species 2 direct pairs	**Y1 SSO1** GenAge 1381 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526 **Y2 SSO2** GenAge 1281 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 unc-64** GenAge 710 **GenAge annotation** **Unannotated** Lifespan: Increase Refs: 10377425	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W1–Y2 (n:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C083 **Surf1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	No gene in this candidate network	**F1 Surf1** GenAge 636 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16172499	**M1 Surf1** GenAge 637 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17210671
Direct pairs: F1–M1 (1:1)				
C025 **SWD1** 2 genes / 2 species 1 direct pair	**Y1 SWD1** GenAge 1418 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 F21H12.1** GenAge 238 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C109 **SWD3** 2 genes / 2 species 1 direct pair	**Y1 SWD3** GenAge 1586 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 wdr-5.1** GenAge 1000 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20555324, 22012258	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:1)				
C118 **TAL1** 2 genes / 2 species 1 direct pair	**Y1 TAL1** GenAge 1449 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19030232	**W1 Y24D9A.8** **tald-1** GenAge 1939, 2213 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24662282, 28355222	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C036 **TDH2** 3 genes / 2 species 2 direct pairs	**Y1 TDH2** GenAge 2078 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 23342062	**W1 gpd-2** GenAge 321 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17673661 **W2 gpd-3** GenAge 1778 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 23144747	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W2–Y1 (n:m)				
C043 **TIF4632** 2 genes / 2 species 1 direct pair	**Y1 TIF4632** GenAge 1661 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043	**W1 ifg-1** GenAge 362 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16720740, 17266679 17266680, 17411345 20700440	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C087 **TOP1** 2 genes / 2 species 1 direct pair	**Y1 TOP1** GenAge 1547 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 19030232	**W1 top-1** GenAge 676 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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	Budding yeast	Nematode	Fruit fly	Mouse
C063 **TPK1** 4 genes / 2 species 3 direct pairs	**Y1 TPK1** GenAge 1502 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043 **Y2 TPK2** GenAge 681 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11000115 **Y3 TPK3** GenAge 1632 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	No gene in this candidate network	**F1 Pka-C1** GenAge 507 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 17369827	No gene in this candidate network
Direct pairs: F1–Y1 (n:m); F1–Y2 (n:m); F1–Y3 (n:m)				
C108 **unc-51** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 unc-51** GenAge 996 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 18219227, 19469880	**F1 Atg1** GenAge 1908 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 25199830	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C091 **VPH1** 3 genes / 2 species 2 direct pairs	**Y1 VPH1** GenAge 1719 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043	**W1 unc-32** GenAge 705 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 10747056 **W2 vha-6** GenAge 715 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17411345	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W2–Y1 (n:m)				
C011 **vps-33.1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 vps-33.1** GenAge 720 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	**F1 car** GenAge 89 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 17435236	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C096 **VPS20** 2 genes / 2 species 1 direct pair	**Y1 VPS20** GenAge 1745 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 22912585	**W1 vps-20** GenAge 764 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C107 **VPS21** 2 genes / 2 species 1 direct pair	**Y1 VPS21** GenAge 1180 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20657825	**W1 rab-5** GenAge 975 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19627265, 23144747	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

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	Budding yeast	Nematode	Fruit fly	Mouse
C086 **VPS24** 2 genes / 2 species 1 direct pair	**Y1 VPS24** GenAge 1330 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 T27F7.1** GenAge 652 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:1)				
C051 **VPS41** 2 genes / 2 species 1 direct pair	**Y1 VPS41** GenAge 1331 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 19030232, 20421943	No gene in this candidate network	**F1 lt** GenAge 419 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17435236	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				
C113 **YIH1** 2 genes / 2 species 1 direct pair	**Y1 YIH1** GenAge 1297 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 impt-1** GenAge 2087 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 27717342	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C125 **YOR1** 2 genes / 2 species 1 direct pair	**Y1 YOR1** GenAge 1677 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 20657593	No gene in this candidate network	**F1 Mrp4** GenAge 1874 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 25474322	No gene in this candidate network
Direct pairs: F1–Y1 (n:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.

Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C065 **YPT7** 2 genes / 2 species 1 direct pair	**Y1 YPT7** GenAge 1208 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043, 18690010 20657825	**W1 rab-7** GenAge 530 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C088 **YUH1** 2 genes / 2 species 1 direct pair	**Y1 YUH1** GenAge 1442 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 ubh-3** GenAge 694 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16673436	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:1)				
C103 **ZWF1** 2 genes / 2 species 1 direct pair	**Y1 ZWF1** GenAge 1412 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20421943	No gene in this candidate network	**F1 Zw** GenAge 883 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18809674	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.